Deception-Aware Retrieval-Augmented Generation for Regional Situation Awareness

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Human and Biological Systems — Warfighter Decision Superiority and Readiness

Problem. Regional analysts must synthesize changing news, public statements, social posts, and field reports. These sources can contradict one another, repeat an outdated claim, or deliberately mislead. Standard retrieval-augmented generation (RAG) can return a fluent update while treating five copied articles as five independent confirmations. After a hurricane affects a coastal port, for example, coordinated posts may falsely report that an access route has reopened, while a later official correction receives little attention. The central failure is not arithmetic: it is losing the origin, timing, and contestation of claims during retrieval and generation.

Central hypothesis. RAG grounded in evolving event records that track source dependence, revisions, and competing claims will make regional situation summaries more accurate and less vulnerable to coordinated deception. This extends event extraction beyond our NSF CAREER setting to adversarial evidence and the reliability of generated situation assessments.

Research plan.

- **Build contest-aware regional event records.** Extract who did what, where, and when from heterogeneous streams while retaining source, timestamp, and confidence for each assertion. Detect corrections, retractions, and likely copying relationships among reports; keep credible competing versions visible rather than merging them into a false consensus.
- **Generate explanations that preserve what is contested.** Retrieve event-linked evidence and generate situation updates with claims marked supported, disputed, or unverified. Adapt fine-grained entity, relation, and factuality feedback from our vision-language work to textual RAG. Train the system to flag coordinated amplification, incompatible timelines, and unsupported certainty, and evaluate when it should request additional independent sources.
- **Create controlled information environments for stress tests.** Generate synthetic regional report streams from event schemas, with known event histories and planted delays, copied sources, omissions, and deceptive claims. Vary the adversary's strategy and information delay to probe failure modes. The resulting data reservoir can support what-if exercises and simulated information-environment scenarios.

Evaluation and expected outcomes. We will use GDELT event streams and public disaster timelines, supplemented with synthetic report sets containing known planted deception. We will measure claim-level support, timeline currency, deception susceptibility, and detection of source dependence. Baselines will include standard RAG and iterative RAG with LLM self-critique.

Naval relevance. For humanitarian assistance and disaster relief in a contested littoral region, a Navy or Marine Corps staff needs an up-to-date picture of which roads, ports, and public services are functioning, even as rumors and manipulated reporting spread. This project targets the program's AI-enhanced RAG, regional situation awareness, decision tools for operations in the information environment, and exercise-supporting information reservoirs.

Prior work and PI qualifications. QA-based event extraction (EMNLP 2020), FaithScore (Findings of EMNLP 2024), and FGAIF (TMLR 2025) provide foundations for event representation and fine-grained feedback. The PI received a 2024 NSF CAREER award and, as a postdoc, led system integration for the RESIN team's DARPA KAIROS delivery system, which achieved top performance among participating teams.